czii-cryo-et-object-identification

3rd Place Solution

Rank	3
Team	ONCE UPON A MOON
Author	soso
Topic ID	561417
Version	Original

Source: https://www.kaggle.com/competitions/czii-cryo-et-object-identification/discussion/561417
Retrieved with Kaggle CLI on 2026-10-21.

Thanks kaggle&host for this interesting competition.
What I like for this competition is that the host give out a baseline(especially for data processing), it is very helpful for people like me who has no knowledge with this domain.
Another reason I join this competition was that I want to writing training code based on the library accelerate(I use pure pytorch before).
Since I am on the top, I will think there is no critical bug in my training code.

Summary

Before I start this competition I thought it was a OD task not a segmentation task untile @hengck23 publish his good notebook.
My solution is based on 3D unet with post processing using cc3d. Cross Entropy loss is used with all the 7 particles.
My best solution is 4 fold(7 KF) average ensemble of backbone res101.
One thing I want to mention is that my best solution has same score(0.783) on both Public and Private LB.

models

I use the code from segmentation_models_pytorch_3d.
The best solution is Unet + resnet101.
I also try different architecture with different backbones, but unet+resnet101 is the best on public LB.
I did not dig much of the code, so mostly default parameters was used for these models.

train

I use EMA because it is easier to handle than SWA, although I remember there is saying SWA is better than EMA.
I train the model with input size (64, 128, 128), but inference with (64, 256, 256), which give 0.001 improvement on public LB.
I use half of the original radius during training, considering how the evaluation score is calculated, which is also best based on my experiment.

Augmentation

It is obvious data augmentation will help a lot for this competition considering the rare data we have.
What I used:

- flip on axis x, y, z
- switch axis x and y
- different algos: "denoised", "wbp", "ctfdeconvolved", "isonetcorrected

- simple copy past
- mixup

TTA

2 tta was used, output is averaged with original:

- flip x, y, z
- rot90 for x, y

ensemble

- 4 fold of 7KF average ensemble

Failures

- Try to pretrain on the external data provided by host
- ensemble unet with different backbone like resnet34 and resnet10

Supplemental Q&A

> **AnnieGo · 2025-02-06T02:56:50.767000**
>
> Congratulations on winning the gold medal! It's impressive that you achieved comparable results to our seven-model ensemble using only four models. Could you kindly share the public and private scores for each of your models?

> **soso · 2025-02-06T03:18:57.740000**
>
> There is one model which has 0.773 public, 0.770 private.

> **Dive Deeper · 2025-02-06T04:27:32.693000**
>
> Congrats @tangtang1999. May I know the improvement of using heavy augmentation like different algo, mixup and copypaste?

> **soso · 2025-02-06T08:38:29.907000**
>
> Unfortunately I did not find the logs(When have too much logs, notebook will become too slow, I deleted some logs 😅).
> What I remember is:
>
> 1. When I increase aug algo prob from 0.5 to 0.75 it did not give me better CV

2. For copy past and mixup both did not improve CV but improve public LB. I only submit one fold(5KF for early experiment), because I am a lazy people
3. For others I do not remember the number but I only keep when have better CV.

Remember for this competition we have too less training data, so we can not always trust the local CV

Leo Yang · 2025-02-11T05:53:30.467000

Hi, thank you very much for publishing the solution!

As far as I am concerned, you "train the model with input size (64, 128, 128), but inference with (64, 256, 256)". May I ask **what effect will it bring up to have different size for training and inference? How should we understand it in this situation** ?

Best
Leo

soso · 2025-02-11T06:07:09.167000

Because it worked in public LB.
I think when split it with small patch there will be more border pixels which is hard for the model to predict!

Leo Yang · 2025-02-12T00:51:19.330000

Thanks for replying. I am very interested by this case. May I ask more questions?

Supposed we change the input size for inference, will the results be affected in a bad way?
For example, what if we input a 2048*2048 pixel resolution image, Can it still segmentation correctly ?

soso · 2025-02-12T01:54:02.697000

1. with bigger input size in inference, it have bigger difference compare to train, this should decrease performance. Remember for a convolution models the receptive field is fixed, when input size bigger than receptive field, there should be no more decrease.
2. with bigger input size, you get more context information, this should increase the performance. Same as above about the receptive field
3. with bigger input size, you have less border pixels, this should increase the performance

So there must be some balance.